Symmetry

Name _____ Date _____ No calculator Version 1

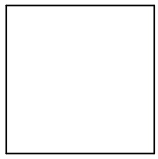
Scan to practise this topic online

KEY WORDS symmetry *(hangarite)* · line of symmetry *(rārangi hangarite)* · mirror line *(rārangi whakaata)*

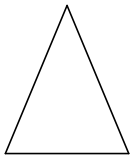
A Lines of symmetry ●○○

> **Remember:** A line of symmetry cuts a shape into two halves that are mirror images. If you fold the shape along the line, the two halves fit exactly on top of each other.

1. Draw all the lines of symmetry on each shape. Then write how many lines of symmetry it has. Some shapes have none.

a) 

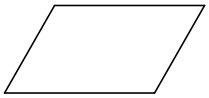
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b) 

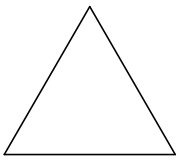
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c) 

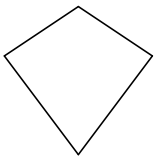
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d) 

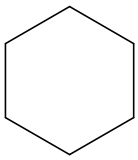
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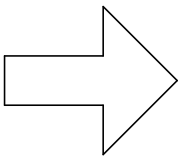
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f) 

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g) 

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h) 

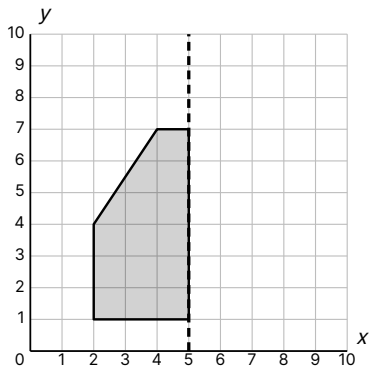
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B Completing a symmetrical shape ●●○

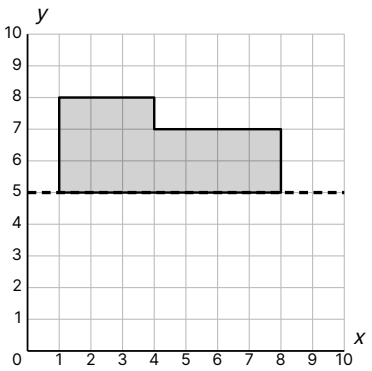
> **Remember:** Each corner of the missing half is the same distance from the mirror line as its partner, but on the other side.

2. Half of a shape is drawn. The dashed line is a line of symmetry. Draw the other half.

a)



b)

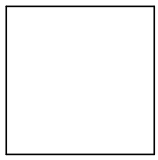


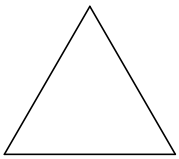
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C Rotational symmetry ●●●

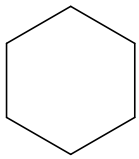
> **Remember:** Turn a shape through one full turn about its centre. The order of rotational symmetry is the number of times it looks exactly the same. A square looks the same 4 times, so its order is 4. A shape that only looks the same once has order 1.

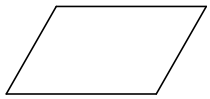
3. Write the order of rotational symmetry of each shape.

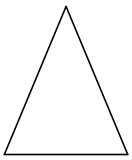
a)  _____

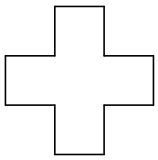
b)  _____

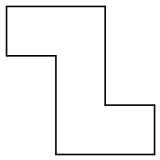
c)  _____

d)  _____

e)  _____

f)  _____

g)  _____

h)  _____

D Both kinds of symmetry ●●●

4. Complete the table.

Shape	Lines of symmetry	Order of rotational symmetry
Square		
Rectangle		
Equilateral triangle		
Parallelogram		
Regular hexagon		

★ Extension

5. A regular hexagon is turned about its centre. Through how many degrees must it turn before it first looks exactly the same?

6. A parallelogram has rotational symmetry of order 2, but it has no lines of symmetry. Explain how you could check each of these with a piece of paper.

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Symmetry

Name _____ Date _____ No calculator Version 2

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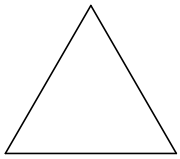
KEY WORDS symmetry *(hangarite)* · line of symmetry *(rārangi hangarite)* · mirror line *(rārangi whakaata)*

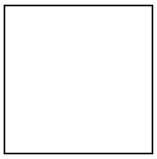
A Lines of symmetry ●○○

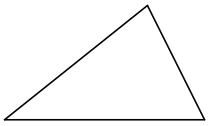
Remember: A line of symmetry cuts a shape into two halves that are mirror images. If you fold the shape along the line, the two halves fit exactly on top of each other.

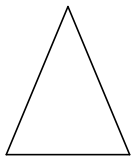
1. Draw all the lines of symmetry on each shape. Then write how many lines of symmetry it has. Some shapes have none.

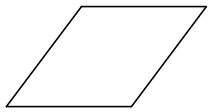
a)  _____

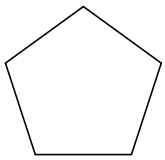
b)  _____

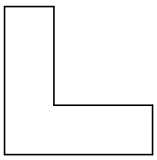
c)  _____

d)  _____

e)  _____

f)  _____

g)  _____

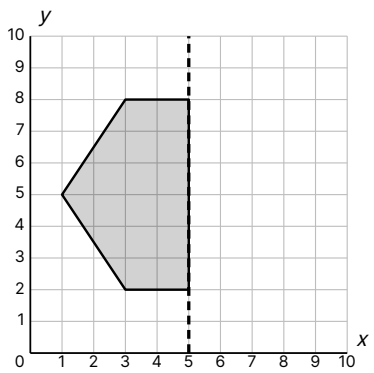
h)  _____

B Completing a symmetrical shape ●●○

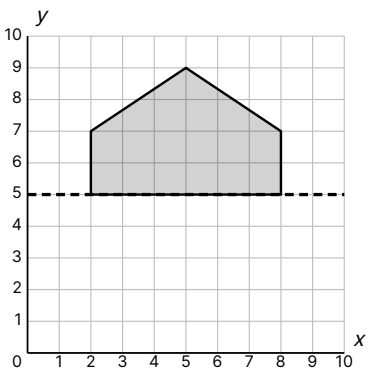
Remember: Each corner of the missing half is the same distance from the mirror line as its partner, but on the other side.

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a)



b)

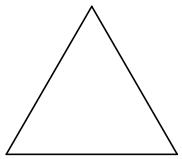


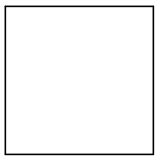
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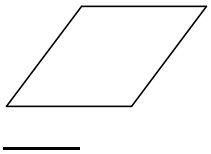
C Rotational symmetry ●●●

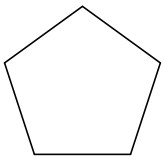
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3. Write the order of rotational symmetry of each shape.

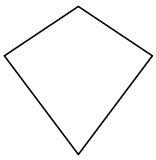
a)  _____

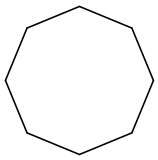
b)  _____

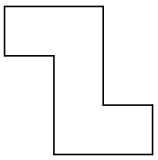
c)  _____

d)  _____

e)  _____

f)  _____

g)  _____

h)  _____

D Both kinds of symmetry ●●●

4. Complete the table.

Shape	Lines of symmetry	Order of rotational symmetry
Rectangle		
Square		
Isosceles triangle		
Rhombus		
Regular pentagon		

★ Extension

5. A regular octagon is turned about its centre. Through how many degrees must it turn before it first looks exactly the same?

6. A parallelogram has rotational symmetry of order 2, but it has no lines of symmetry. Explain how you could check each of these with a piece of paper.

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